

FleetGuard Industrial IoT Monitoring System

Engineering Report

Nova IoT Systems – Embedded Systems Internship Assignment

Prepared By: Nitin Dhanraj Birajdar

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1. Executive Summary

FleetGuard Industrial IoT Monitoring System is an intelligent edge monitoring solution designed for refrigerated transport containers carrying pharmaceuticals and temperature-sensitive products.

Traditional manual monitoring methods introduce several operational challenges, including delayed issue detection, product spoilage, regulatory compliance risks, and the absence of historical data for analysis.

This project proposes a scalable Industrial IoT edge device capable of continuously monitoring environmental conditions, processing data locally, generating alerts, storing historical records, and supporting future cloud integration.

The solution is designed as a production-oriented Minimum Viable Product (MVP) that can eventually scale from a few devices to thousands of deployed units.

2. Problem Analysis

Current challenges:

- Delayed detection of temperature excursions
- Product spoilage risks
- Regulatory compliance concerns
- High manual operational effort
- Lack of historical traceability

Project objectives:

- Monitor temperature and humidity continuously
 - Detect abnormal conditions
 - Generate local alerts
 - Store historical events
 - Support future cloud connectivity
 - Design for scalability
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3. Functional Requirements

Mandatory Features

- Temperature monitoring
- Humidity monitoring

Additional Features

- Door status detection
- Light detection
- Vibration detection
- Battery voltage monitoring

Local Intelligence

System states:

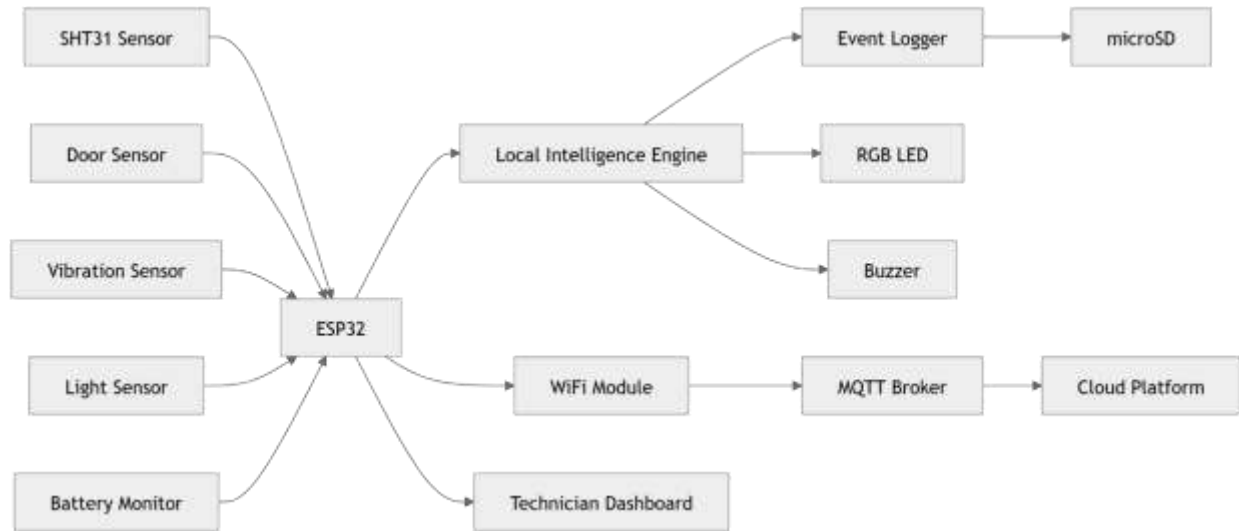
- NORMAL
- WARNING
- CRITICAL

Event Logging

Store:

- Sensor failures
 - Threshold violations
 - Device startup events
 - Configuration changes
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4. System Architecture



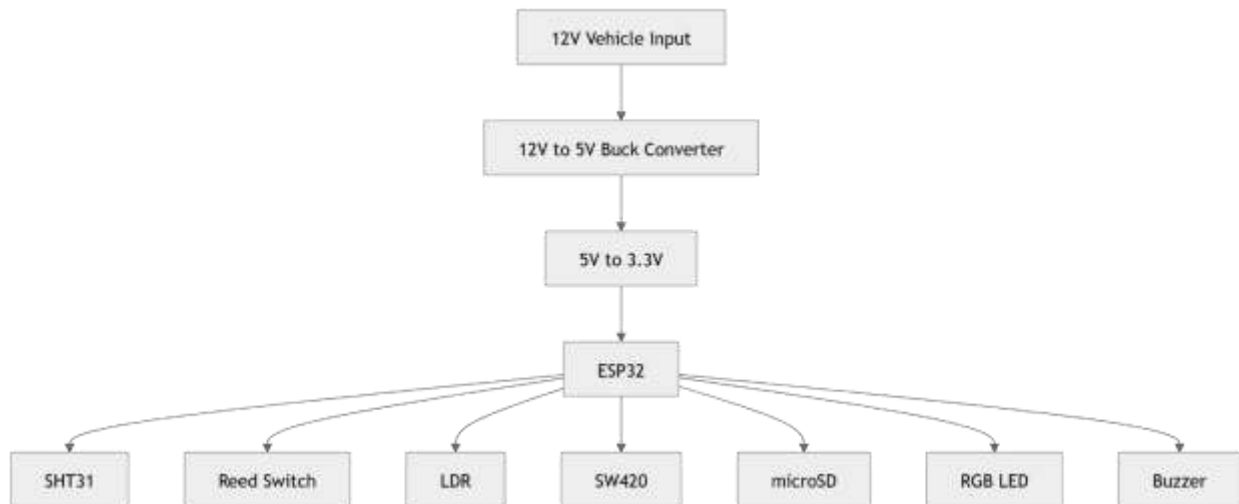
The ESP32 acts as the central controller responsible for edge processing and decision making.

5. Hardware Design

Components

Component	Purpose
ESP32 DevKit V1	Main controller
SHT31	Temperature and humidity
Reed Switch	Door status
LDR	Light detection
SW420	Vibration detection
MicroSD	Event storage
RGB LED	Visual alerts
Buzzer	Audible alerts

Hardware Block Diagram:



6. Firmware Architecture

Modules:

1. Sensor Manager

Responsibilities:

- Initialize sensors
- Read sensors
- Retry failed sensors

2. Validation Engine

Responsibilities:

- Verify ranges
- Detect invalid data

3. Classification Engine

Responsibilities:

- Determine system state

4. Alert Manager

Responsibilities:

- Activate LEDs
- Trigger buzzer

5. Event Logger

Responsibilities:

- Save timestamps
- Store events

6. Configuration Manager

Responsibilities:

- Load thresholds
 - Store settings
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7. Threshold Strategy

Temperature

- Normal: 2°C to 8°C
- Warning: 0°C to 2°C or 8°C to 10°C
- Critical: Below 0°C or Above 10°C

Humidity

- Normal: 40% to 60%
 - Warning: 35% to 40% or 60% to 70%
 - Critical: Below 35% or Above 70%
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8. Reliability and Scalability

Reliability mechanisms:

- Sensor retry logic
- Event persistence
- Future watchdog support
- Local operation without internet

Scalability:

10 Devices: Direct Wi-Fi

100 Devices: Central MQTT Broker

9. Security Considerations

- Device Identity: Unique Device ID
 - Future Secure Communication: MQTT over TLS
 - Firmware Security: Digital firmware signing
 - Configuration Security: Protected device settings
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10. Risks and Future Improvements

Risks

Sensor failure

- Mitigation: Retry mechanism, Power interruption.
- Mitigation: Persistent storage ,Communication loss.
- Mitigation: Local processing , False alarms .
- Mitigation: Multiple sample validation.

Future Improvements

- GPS tracking
 - Cellular connectivity
 - OTA updates
 - Predictive maintenance
 - Analytics dashboard
 - Sensor redundancy
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11. Conclusion

The FleetGuard Industrial IoT Monitoring System provides a practical and scalable solution for monitoring refrigerated transport containers.

The design prioritizes reliability, maintainability, cost effectiveness, and future scalability while remaining suitable for deployment as an Industrial IoT edge device.

This architecture can serve as a foundation for large-scale fleet deployments.